Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Non-linear representations in quantum mechanics

Files.

- File: `briefs/NonLinearQuantum/NonLinearQuantum.lean`
- Library: `NonLinearQuantum`
- Imports: `Mathlib.Analysis.InnerProductSpace.Adjoint`, `Mathlib.Analysis.InnerProductSpace.Calculus`, `Mathlib.Analysis.SpecialFunctions.ExpDeriv`, `Mathlib.LinearAlgebra.Projectivization.Basic`
- File: `briefs/NonLinearQuantum/NonLinearQuantum/DensityOperators.lean`
  Density operators and entropy on $m(H)$, in a module of its own, so that the comparator lane for the main file is unchanged.
- Imports: `NonLinearQuantum`, `ForPhyslib.QuantumMechanics.UnitaryEquivalence`

Setting. E is any complex inner product space, $v, w \in E$ are orthonormal (`Orthonormal ℂ ![v, w]`), and

$$m(x) = e^{i\theta\, r(x)}\, x, \qquad r(x) = p(v \mid x) = \frac{|\,\langle v|x\rangle\,|^2}{\|v\|^2\|x\|^2}$$

(`nlMap` v θ). For $x = av + bw$ this is the brief's eq. (2) (`nlMap_smul_add_smul`), and in Lean $m(0) = 0$ holds without a case split, since $0/0 = 0$. The brief's H is the case $\dim E = 2$. The Fubini–Study distance is $d(x, y) = \arccos(|\,\langle x|y\rangle\,|/(\|x\|\,\|y\|))$ (`fsDist`), and the ray $[x]$ of $x \neq 0$ is its point `Projectivization.mk ℂ` x `hx` of the projective space.

Results.

Brief	Lean
Prop. 1, eq. (2)	`nlMap_smul_add_smul`; in dimension two it determines m: `eq_nlMap`
Prop. 1, m bijective, inverse for $-\theta$	`nlHomeomorph` (a homeomorphism), `nlMap_neg_nlMap`, `nlMap_nlMap_neg`
Prop. 1, $[m(x)] = [x]$	`mk_nlMap`; `nlMap_mem_ray`: $m(x) = c\,x$ with $\lvert c\rvert = 1$
Prop. 1, Fubini–Study distance preserved	`fsDist_nlMap`, from `inner_nlMap`
Prop. 1, $m(v+w) \neq m(v) + m(w)$	`nlMap_not_add`, `not_isLinearMap_nlMap`, for $\theta \in (0, 2\pi)$
Prop. 1, all together	`nlMap_spec`
Cor. 3, A self-adjoint and unitary	`swapOp`, `isSelfAdjoint_swapOp`, `swapOp_mem_unitary` (dimension two)
Cor. 3, the values of $\tilde{A}$	`transport_swapOp_left`, `transport_swapOp_right`, `transport_swapOp_add`
Cor. 3, $\tilde{A}$ not additive	`not_map_add_transport_swapOp`
Thm. 4, proof: Born rule $= \cos^2 d$	`born_eq_cos_sq_fsDist`
Thm. 4, proof: Born rule preserved	`born_nlMap`
(correction) density operators	`not_map_add_transport_projOp`: $m\,\lvert v\rangle\langle v\rvert\,m^{-1}$ is not additive
Thm. 4, proof: $m(H)$	`Twisted`, `toTwisted` (identity of the carrier)
same scalars, hence same rays	`toTwisted_smul`
same Born rule	`born_toTwisted`
different addition	`not_map_add_toTwisted`
m a linear isometry onto $m(H)$	`twistedIsometry`
all together	`exists_second_linear_structure`
a separable Hilbert space	`instCompleteSpaceTwisted`, `instSeparableSpaceTwisted`, `finrank_twisted`
states unchanged	`mk_twistedIsometry`: the ray of $m(x)$ in $m(H)$ is that of x
density operators	`transportOp`: $\rho \mapsto m\rho m^{-1}$; `densityEquiv`: a bijection of the density operators
spectra and entropy kept	`spectrum_transportOp`, `vonNeumannEntropy_densityEquiv`
on the carrier, not additive	`transportOp_toTwisted`, `not_map_add_transportOp_projOp`
QST:H on $m(H)$, together	`qst_twisted`
Conclusion, inner product not conserved	`inner_nlMap_ne`
(extra) smoothness	`contDiffAt_nlMap`: smooth at every $x \neq 0$; `not_differentiableAt_nlMap_zero`: not differentiable at 0
(extra) Born rule depends only on rays	`born_smul`: $p(cx \mid dy) = p(x \mid y)$ for $c, d \neq 0$
(extra) dimension one	`map_add_of_map_smul`: maps commuting with scalars are additive

Differences from the brief.

- *Dimension.* The brief states Proposition 1 and Corollary 3 for a two-dimensional H. In Lean E is any complex inner product space with an orthonormal pair, so any dimension ≥ 2,

finite or not. Dimension two is used only where it is needed: that eq. (2) determines m (`eq_nlMap`), and that A is unitary (`swapOp_mem_unitary`). Completeness is assumed only for self-adjointness and unitarity, which are stated in the C*-algebra $E \to_L E$.

- *Corollary 3.* The brief's $A(av+bw) = bv+aw$ is written as $Ax = \langle w|x\rangle\ v + \langle v|x\rangle\ w$ (`swapOp`); in dimension two the two agree.

- *Fubini–Study distance.* `fsDist` is defined for all x, y; for $x = 0$ or $y = 0$ it is $\arccos 0$. Preservation holds for all x, y, so the brief's restriction to $x, y \neq 0$ is not needed.

- *Theorem 4.* The brief now separates mathematical prerequisites (Sec. 2) from the physical conditions and Theorem 4. Following that split, Theorem 4 is not stated in Lean; it is checked by reading, against the Lean results above. The argument is valid under the premise, stated in the abstract, that a map which preserves the Born rule preserves "all the physics", so that moving every physical object over through m gives a valid representation. Each step that is mathematical is checked: the states (`mk_nlMap`), the Born rule as $\cos^2$ of the Fubini–Study distance (`born_eq_cos_sq_fsDist`, `fsDist_nlMap`), that superposition does not move over to vector addition (`nlMap_not_add`), and that self-adjoint and unitary operators move over to non-linear maps (Corollary 3). See the problems below for the two places where the wording needs a correction.

Problems found in the brief.

- The proof of Theorem 4 says that density operators "will be left unchanged together with their entropy". As operators they are not: moving $|v\rangle\langle v|$ over through m gives $\tilde\rho = m\,|v\rangle\langle v|\,m^{-1}$ with $\tilde\rho(v) = v$, $\tilde\rho(w) = 0$ and $\tilde\rho(v+w) = e^{\imath\theta/2}v$, which is not additive (`not_map_add_transport_projOp`). Keeping the original ρ instead would not fit with the moved-over observables, since $\mathrm{tr}(\rho\tilde A)$ is not defined for a non-linear $\tilde A$. What is true is that pure states, as rays, are unchanged (`mk_twistedIsometry`), and that $\rho \mapsto m\rho m^{-1}$ is a bijection from the density operators of H onto those of $m(H)$ (`densityEquiv`) that keeps spectra and entropy (`spectrum_transportOp`, `vonNeumannEntropy_densityEquiv`); $m(H)$ is a separable Hilbert space like H (`qst_twisted`). QMLIN does not mention density operators, so the conclusion is not affected; only the step "QST:H remains valid on $m(H)$" needs rewording.

- QMLIN asks that superpositions, observables and evolution use "the ordinary linear structure of that Hilbert space". The ordinary structure of H and the structure carried over along m have exactly the same rays and Born rule (`toTwisted_smul`, `born_toTwisted`), and with the carried-over structure QMLIN holds: superposition is its addition, and $\tilde A$ is linear for it (`twistedIsometry`). So Theorem 4 needs "that Hilbert space" to mean H with its given operations. The proof reads it so, but a sentence saying it would answer the objection that $m(H)$ is a linear representation in disguise.

- Numbering: the results are Proposition 1, Corollary 3 and Theorem 4, and the only numbered equation is (2). The proof of Theorem 4 refers to "m as defined by 1" and "by 1 and 3".

- The Fubini–Study distance, and so the Born rule, is preserved by every map that sends each vector into its own ray (`born_smul`); any $x \mapsto \varphi([x])\,x$ with $\varphi\colon P(H) \to \mathbb{C}\setminus\{0\}$ would do. What the proof needs from the particular m is that it is not additive.